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1. In step 2, what is the utility of converting x to zero mean and unit variance?
Ans: It scales all the values to 1 and 0, making it easier to converge and faster to learn

2. In step 3, what is the need to use seq() function?
Ans: seq generates a sequence of given from and to numbers, with a gap of ‘by’ value.

3. What are your observations in step 4?
Ans: As the value of n increases, the gaussian distribution curve fits the histogram perfectly.

4. In step 7, why is correlation used to identify best two features? Will this always give good results?
Ans: Columns 9 and 8 give the two best correlation. No, they will not always give the best results.

5. Mysteriously, both the data and the residuals from linear Regression can be fitted used Gaussian distribution. Is this by chance? Search the internet to answer this question in depth.
Ans: According to the central limit theorem, when you add up a lot of small, random effects, the results tend to be normal, even if the original factors aren’t. Hence residuals end up look like a normal distribution.